

AMISR Implicit Neural Representation for Ionospheric Plasma Density Reconstruction

Technical Summary for Discussion with Gonzalo Ruz

Internal research notes

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1 Purpose of this document

This document summarizes the current state of the AMISR implicit neural representation project, with emphasis on the methodological decisions made so far, the mathematical formulation, the practical code changes, and the path forward. The intended audience is a machine-learning expert who can comment on the modeling choices, regularization strategy, temporal windowing, and possible next architectures.

The project began as a simple application of SIREN-style neural fields to radar-derived plasma density slices. It has now evolved into a more structured physics-aware machine-learning workflow. The current method combines:

- direct use of AMISR HDF5 data products;
- a three-dimensional coordinate input, namely horizontal position and time;
- low-frequency SIREN activations as a spectral prior;
- local temporal windows motivated by plasma correlation time;
- second-derivative curvature regularization in space and time;
- collocation-based derivative penalties;
- reference-based adaptive scaling of regularization weights;
- per-window prediction and diagnostic plotting;
- an upcoming window-merging step to build a continuous time-dependent reconstruction.

The scientific motivation is that a neural network can easily fit sparse radar data while inventing unrealistic structure between beams. Therefore, the interpolation itself must be constrained by physical and statistical assumptions. The current method uses smoothness, spectral bias, local temporal context, and conservative visualization masks to avoid over-interpreting unsupported regions.

2 Acronyms and terminology

Acronym term	or	Meaning
AMISR		Advanced Modular Incoherent Scatter Radar. In this project, the term refers to the radar data product being used for ionospheric plasma density measurements.
ISR		Incoherent Scatter Radar. A radar technique used to estimate ionospheric plasma parameters such as electron density, ion/electron temperature, and line-of-sight velocity.
INR		Implicit Neural Representation. A neural network representation of a continuous signal or field, where coordinates are mapped directly to field values.
SIREN		Sinusoidal Representation Network. An INR architecture using sinusoidal activation functions, typically written as $\sin(\omega_0 z)$.

MLP	Multi-Layer Perceptron. A feed-forward neural network.
HDF5	Hierarchical Data Format version 5. A file format used to store large structured scientific datasets.
PINN	Physics-Informed Neural Network. A neural-network method that includes differential-equation residuals or derivative constraints in the loss. Our method is PINN-like in using collocation derivative losses, but the current derivative terms are regularizers, not a full physical PDE.
PDE	Partial Differential Equation. A differential equation involving derivatives with respect to multiple variables, such as space and time.
EMA	Exponential Moving Average. A smoothed running average used to stabilize noisy quantities such as loss values.
RMSE	Root Mean Square Error. A square-root version of mean squared error, often easier to interpret physically.
FoV	Field of View. The spatial region observed by the radar geometry.
Hessian	Matrix of second partial derivatives. For a two-dimensional function $f(x, y)$, the Hessian contains f_{xx} , f_{xy} , f_{yx} , and f_{yy} .
Laplacian	Trace of the Hessian. In two spatial dimensions, $\nabla^2 f = f_{xx} + f_{yy}$. This is not identical to the full Hessian penalty currently used.
Collocation points	Artificially sampled points in the domain where derivative regularization terms are evaluated, even if no radar measurement exists at those exact coordinates.
Window	A local temporal subset of the radar data used to train an independent INR model.
Stride	The number of records by which the window start index advances between consecutive trained windows.
Cadence	The time separation between consecutive radar records, for example 5 minutes or 2 minutes.

3 Scientific objective

The target field is the electron density of the ionosphere, expressed in logarithmic form. The model represents this field as a continuous function of horizontal position and time:

$$f_{\theta}(x, y, t) \approx \log_{10} N_e(x, y, t). \quad (1)$$

Here x and y are horizontal coordinates, typically in kilometers, t is time, N_e is electron density, and θ denotes the trainable neural-network parameters.

The current implementation focuses on fixed-altitude slices. A representative altitude band is centered near

$$h_0 \approx 330 \text{ km}, \quad (2)$$

with a finite altitude half-width around that central height. In other words, the current field

is not yet the full four-dimensional field $N_e(x, y, z, t)$. Instead, we work with a height-filtered representation:

$$f_\theta(x, y, t; h_0, \Delta h) \approx \log_{10} N_e(x, y, t \mid h \in [h_0 - \Delta h, h_0 + \Delta h]). \quad (3)$$

The near-term goal is not simply to fit the measured radar points. The goal is to infer a physically plausible plasma morphology between sparse radar samples. This distinction is essential. A high-capacity neural network can interpolate sparse radar measurements in many ways, including ways that are not physically plausible. Therefore, the learned representation must be constrained by domain knowledge.

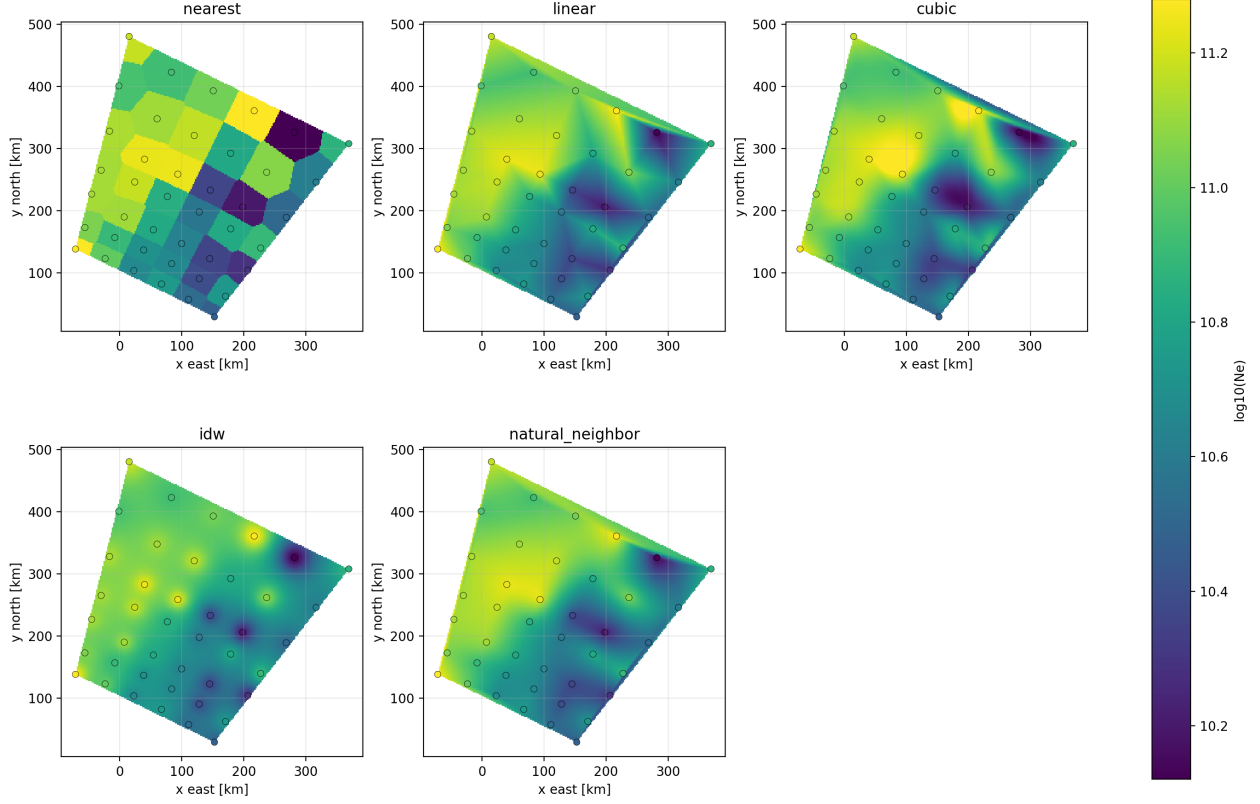


Figure 1: Interpolation examples that are purely mathematical

4 Data representation

For a selected altitude band and a selected temporal window, the data consist of samples

$$\mathcal{D}_w = \{(x_i, y_i, t_i, u_i)\}_{i=1}^{N_w}, \quad (4)$$

where

$$u_i = \log_{10} N_{e,i}. \quad (5)$$

The subscript w denotes a specific time window. The number of samples in that window is N_w . Each sample is derived from a radar beam and a time record, after selecting data within the altitude band.

The network input is the normalized coordinate vector

$$\mathbf{r}_i = \begin{bmatrix} \tilde{x}_i \\ \tilde{y}_i \\ \tilde{t}_i \end{bmatrix} \in \mathbb{R}^3. \quad (6)$$

A typical min-max normalization to $[-1, 1]$ is

$$\tilde{x}_i = 2 \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} - 1, \quad (7)$$

$$\tilde{y}_i = 2 \frac{y_i - y_{\min}}{y_{\max} - y_{\min}} - 1, \quad (8)$$

$$\tilde{t}_i = 2 \frac{t_i - t_{\min}}{t_{\max} - t_{\min}} - 1. \quad (9)$$

The target can also be normalized. If $u_i = \log_{10} N_{e,i}$, then one possible normalized target is

$$\tilde{u}_i = 2 \frac{u_i - u_{\min}}{u_{\max} - u_{\min}} - 1. \quad (10)$$

The model is trained in normalized coordinates and normalized target space, then predictions are denormalized for plotting and physical interpretation.

5 Current neural representation

5.1 Coordinate-based implicit representation

The current model is a coordinate-based neural field. Instead of predicting a value on a fixed grid, the neural network maps a continuous coordinate to a field value:

$$f_\theta : \mathbb{R}^3 \rightarrow \mathbb{R}, \quad (11)$$

$$(\tilde{x}, \tilde{y}, \tilde{t}) \mapsto \tilde{u}. \quad (12)$$

Equivalently,

$$\tilde{u} = f_{\theta}(\tilde{x}, \tilde{y}, \tilde{t}). \quad (13)$$

After denormalization, this becomes an estimate of $\log_{10} N_e$.

5.2 SIREN activation

The baseline architecture is a SIREN-style MLP. A standard hidden layer can be written as

$$\mathbf{h}_{\ell} = \sin(\omega_{\ell} (W_{\ell} \mathbf{h}_{\ell-1} + \mathbf{b}_{\ell})), \quad (14)$$

where ω_{ℓ} controls the frequency scale of the sinusoidal activation. For the first hidden layer,

$$\mathbf{h}_1 = \sin(\omega_0^{(1)} (W_1 \mathbf{r} + \mathbf{b}_1)), \quad (15)$$

and for subsequent layers,

$$\mathbf{h}_{\ell} = \sin(\omega_0^{(h)} (W_{\ell} \mathbf{h}_{\ell-1} + \mathbf{b}_{\ell})). \quad (16)$$

The output layer is typically linear:

$$f_{\theta}(\mathbf{r}) = W_L \mathbf{h}_{L-1} + \mathbf{b}_L. \quad (17)$$

5.3 Low-frequency SIREN as a spectral prior

A major methodological decision has been to use low-frequency SIREN settings. In the original SIREN formulation, relatively large frequency scales are often used, especially in the first layer. For sparse radar plasma reconstruction, this can allow high-frequency oscillations that fit the data but look physically implausible between beams.

The working empirical observation is:

$$\omega_0 \text{ too large} \Rightarrow \text{oscillatory or visually unphysical plasma morphology}, \quad (18)$$

$$\omega_0 \text{ smaller} \Rightarrow \text{smoother and more physically plausible plasma morphology}. \quad (19)$$

This is interpreted as a spectral prior. The network still has nonlinear representational capacity, but the low-frequency setting biases the solution toward large-scale structures. This is consistent

with the expectation that the ionospheric plasma density field should not contain arbitrary beam-to-beam ringing at scales unsupported by the radar geometry.

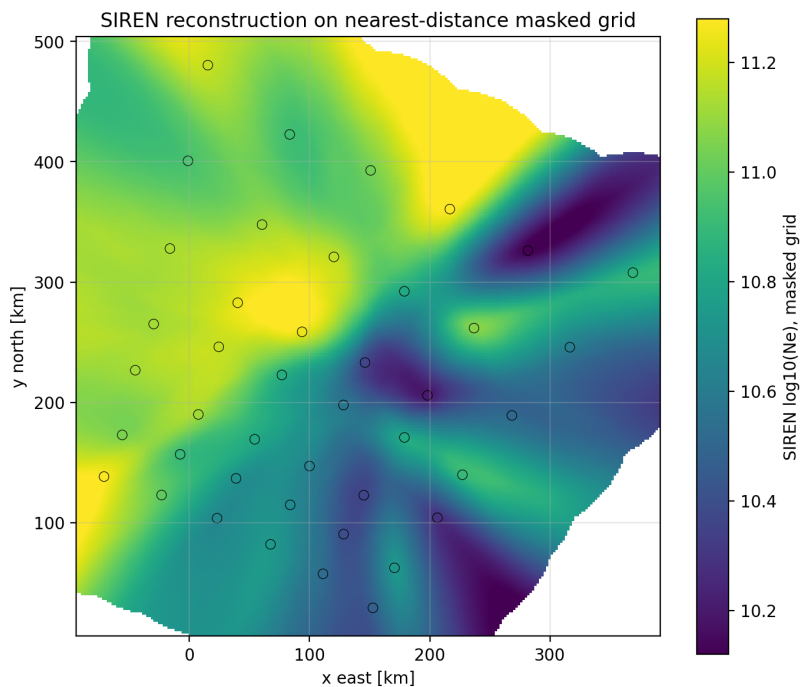


Figure 2: ReLu activation.

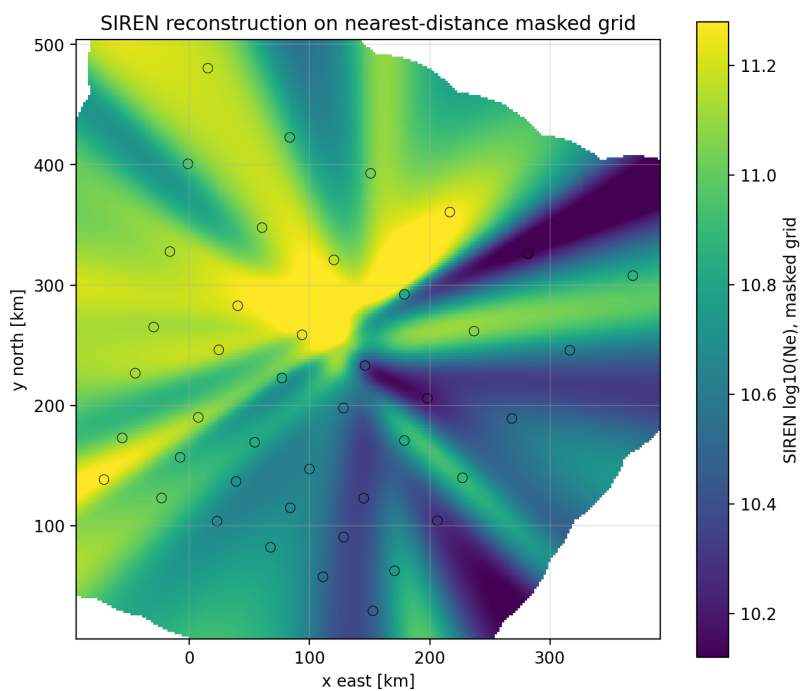


Figure 3: Tanh activation.

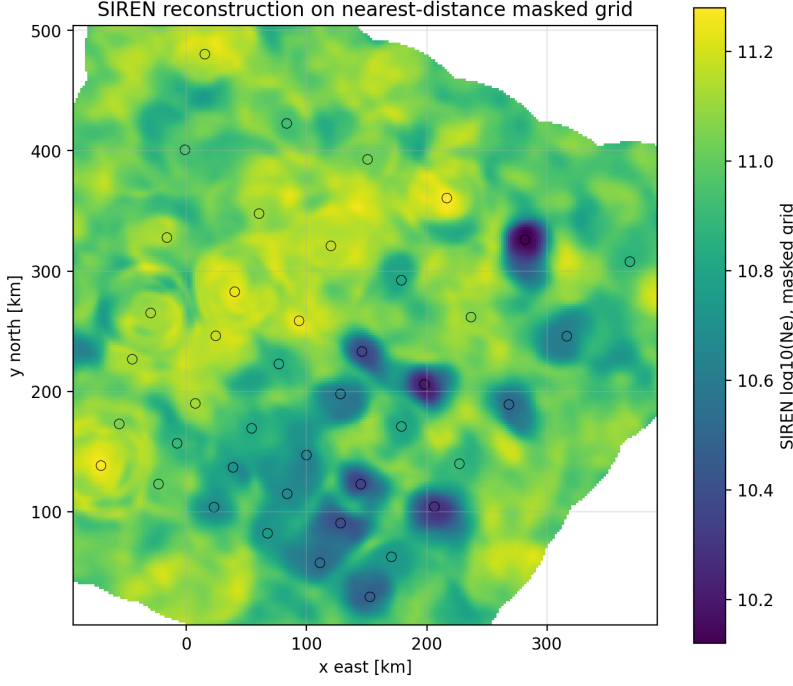


Figure 4: High frequency SIREN

6 Loss function: from interpolation to physics-aware regularization

6.1 Total loss

The current training loss is a weighted sum of a data-fitting term and curvature regularization terms:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{xy}\mathcal{L}_{xy} + \lambda_t\mathcal{L}_t. \quad (20)$$

Here $\mathcal{L}_{\text{data}}$ enforces agreement with measured radar data, $\mathcal{L}_{xy}$ controls horizontal spatial curvature, $\mathcal{L}_t$ controls temporal curvature, and λ_{xy} , λ_t are the corresponding regularization weights.

6.2 Data-fitting loss

The data loss is the mean squared error between the model and the measured normalized target values:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_w} \sum_{i=1}^{N_w} (f_{\theta}(\tilde{x}_i, \tilde{y}_i, \tilde{t}_i) - \tilde{u}_i)^2. \quad (21)$$

In unnormalized notation, this corresponds to

$$\mathcal{L}_{\text{data}} \propto \frac{1}{N_w} \sum_{i=1}^{N_w} (f_{\theta}(x_i, y_i, t_i) - \log_{10} N_{e,i})^2. \quad (22)$$

The proportionality sign is used because the actual implementation typically computes the loss in normalized target space.

6.3 Why not only data loss?

If only $\mathcal{L}_{\text{data}}$ is used, a flexible INR can fit the radar measurements while creating unsupported structures between radar beams. In sparse inverse problems, many functions can pass through the measured points. A useful reconstruction must choose among those functions using priors.

Therefore, the loss is not intended to produce the lowest possible data error at any cost. Instead, the goal is to fit the measured data adequately while selecting the smoothest physically plausible field compatible with those measurements.

7 Curvature regularization

7.1 Why gradient regularization was rejected

An earlier idea was to penalize first derivatives, for example

$$\mathcal{L}_{\nabla} = \mathbb{E}_{(x,y,t) \sim \Omega} \left[\|\nabla f_{\theta}(x, y, t)\|^2 \right]. \quad (23)$$

In expanded form, for spatial derivatives this would include

$$\|\nabla_{xy} f_{\theta}\|^2 = \left(\frac{\partial f_{\theta}}{\partial x} \right)^2 + \left(\frac{\partial f_{\theta}}{\partial y} \right)^2. \quad (24)$$

The problem is that first-derivative regularization discourages gradients themselves. That is often too restrictive for ionospheric plasma, because real plasma density can have large-scale gradients. The physical problem is not that gradients exist. The problem is that unsupported rapid bending, ringing, or oscillatory structure may be introduced between sparse measurements.

Thus, the regularization was changed from first derivatives to second derivatives.

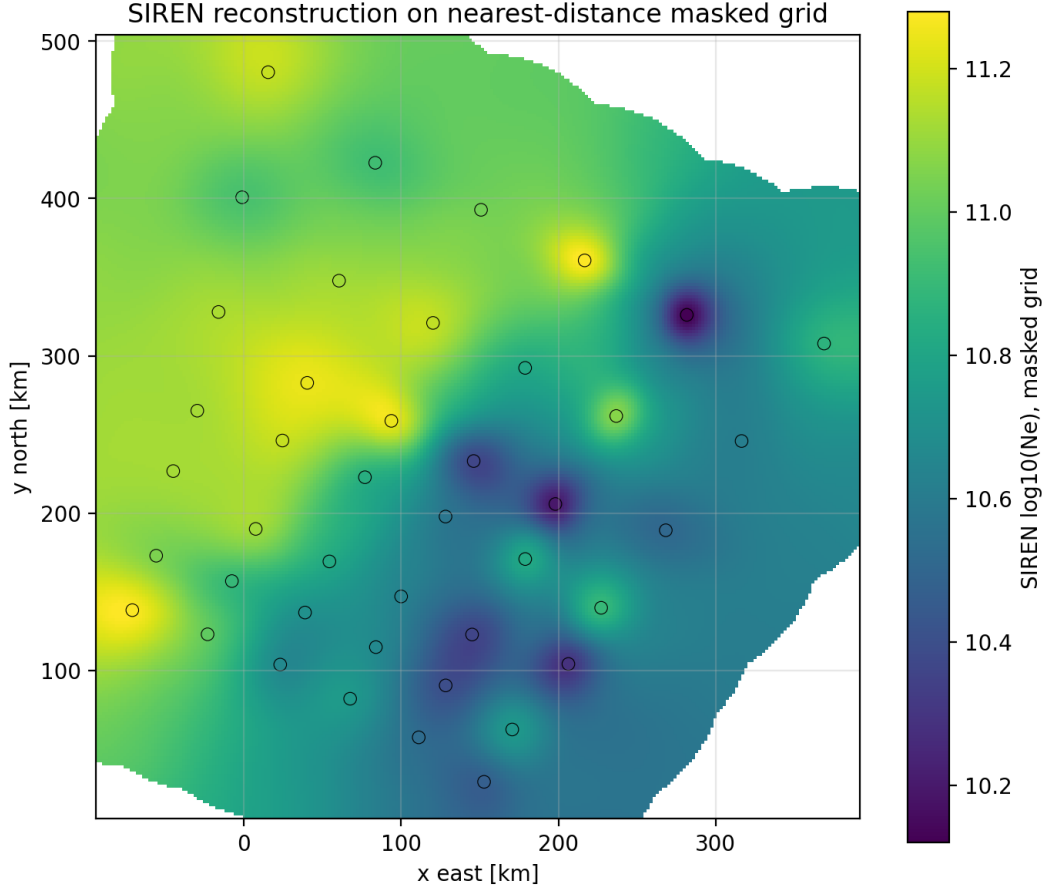


Figure 5: Gradient loss too strong

7.2 Horizontal Hessian

For a function $f_\theta(x, y, t)$, the horizontal Hessian at fixed time is

$$H_{xy}f_\theta = \begin{bmatrix} \frac{\partial^2 f_\theta}{\partial x^2} & \frac{\partial^2 f_\theta}{\partial x \partial y} \\ \frac{\partial^2 f_\theta}{\partial y \partial x} & \frac{\partial^2 f_\theta}{\partial y^2} \end{bmatrix} \quad (25)$$

$$= \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}. \quad (26)$$

Assuming sufficient smoothness, $f_{xy} = f_{yx}$. The squared Frobenius norm of the Hessian is

$$\|H_{xy}f_\theta\|_F^2 = f_{xx}^2 + f_{xy}^2 + f_{yx}^2 + f_{yy}^2 \quad (27)$$

$$= f_{xx}^2 + 2f_{xy}^2 + f_{yy}^2. \quad (28)$$

This is the quantity currently used for the horizontal curvature penalty.

7.3 Spatial curvature loss

The spatial curvature loss is evaluated at collocation points sampled from the reconstruction domain:

$$\mathcal{L}_{xy} = \mathbb{E}_{(x,y,t) \sim \Omega_w} \left[\|H_{xy} f_\theta(x, y, t)\|_F^2 \right] \quad (29)$$

$$= \mathbb{E}_{(x,y,t) \sim \Omega_w} [f_{xx}^2 + 2f_{xy}^2 + f_{yy}^2]. \quad (30)$$

This should be described as a horizontal Hessian curvature penalty, not simply as a Laplacian penalty.

7.4 Distinction from the Laplacian

The two-dimensional horizontal Laplacian is

$$\nabla_{xy}^2 f_\theta = \frac{\partial^2 f_\theta}{\partial x^2} + \frac{\partial^2 f_\theta}{\partial y^2} \quad (31)$$

$$= f_{xx} + f_{yy}. \quad (32)$$

A pure Laplacian penalty might use

$$\mathcal{L}_\Delta = \mathbb{E} \left[(f_{xx} + f_{yy})^2 \right]. \quad (33)$$

This is not the same as the current penalty. The current penalty uses the full horizontal Hessian norm and includes the mixed derivative f_{xy} . It penalizes curvature in all horizontal directions, not only the trace of the Hessian. This distinction is important when explaining the method.

7.5 Temporal curvature loss

The temporal curvature loss is

$$\mathcal{L}_t = \mathbb{E}_{(x,y,t) \sim \Omega_w} \left[\left(\frac{\partial^2 f_\theta}{\partial t^2} \right)^2 \right] \quad (34)$$

$$= \mathbb{E}_{(x,y,t) \sim \Omega_w} [f_{tt}^2]. \quad (35)$$

This discourages rapid temporal bending but does not forbid a steady trend in time. For example, a locally linear evolution

$$f_\theta(x, y, t) = a(x, y)t + b(x, y) \quad (36)$$

has

$$f_{tt} = 0, \quad (37)$$

and is therefore not penalized by $\mathcal{L}_t$. This is desirable because a large-scale monotonic increase or decrease in density may be physically reasonable.

7.6 Interpretation

The curvature penalties allow the model to represent gradients while discouraging rapid changes in those gradients. In one dimension, a linear function

$$f(x) = ax + b \quad (38)$$

has

$$f_x = a, \quad (39)$$

$$f_{xx} = 0. \quad (40)$$

A first-derivative penalty would penalize a , but a second-derivative penalty would not. This is the key reason curvature regularization is better aligned with the plasma physics intuition: gradients are allowed; unnecessary bending is suppressed.

8 Collocation points and domain support

8.1 Collocation sampling

The derivative regularization is not evaluated only at measured radar points. Instead, the method samples collocation points in the radar-supported spatiotemporal domain:

$$(x_c, y_c, t_c) \sim \Omega_w. \quad (41)$$

The empirical curvature losses are then approximated by Monte Carlo averages:

$$\mathcal{L}_{xy} \approx \frac{1}{N_c} \sum_{j=1}^{N_c} [f_{xx}(x_j^c, y_j^c, t_j^c)^2 + 2f_{xy}(x_j^c, y_j^c, t_j^c)^2 + f_{yy}(x_j^c, y_j^c, t_j^c)^2], \quad (42)$$

$$\mathcal{L}_t \approx \frac{1}{N_c} \sum_{j=1}^{N_c} f_{tt}(x_j^c, y_j^c, t_j^c)^2. \quad (43)$$

Here N_c is the number of collocation points. Increasing N_c improves the sampling of the curvature estimate, but it also increases training cost because second derivatives require expensive automatic differentiation.

8.2 Why collocation points are needed

If curvature were evaluated only at measured points, the model could still behave poorly between radar beams. Collocation points make the prior act throughout the reconstruction domain. This is similar in spirit to PINNs, but the current derivative terms are smoothness priors rather than residuals of a known physical PDE.

8.3 Support masking for visualization

Predictions are not equally trustworthy everywhere on a rectangular plotting grid. The radar only supports certain regions of the (x, y) plane. To avoid over-interpreting unsupported regions, a nearest-distance mask is applied in plotting.

Let $\mathcal{P}_t$ be the set of measured radar locations at time t :

$$\mathcal{P}_t = \{(x_i, y_i) : t_i = t\}. \quad (44)$$

For a grid point (x, y) , define its distance to the nearest measured point as

$$d_{\min}(x, y; t) = \min_{(x_i, y_i) \in \mathcal{P}_t} \sqrt{(x - x_i)^2 + (y - y_i)^2}. \quad (45)$$

A plot mask can then be defined by

$$M(x, y; t) = \begin{cases} 1, & d_{\min}(x, y; t) \leq R_{\text{support}}, \\ 0, & d_{\min}(x, y; t) > R_{\text{support}}. \end{cases} \quad (46)$$

where R_{support} is chosen from the typical radar-beam spacing. This does not change the trained model; it only makes the visualization more conservative.

9 Reference-based regularization weights

9.1 Problem with fixed raw lambda values

A raw weighted loss such as Eq. (20) requires choosing λ_{xy} and λ_t . The difficulty is that raw curvature loss magnitudes can change across windows, cadence, normalization, and training stage. Therefore, a fixed value such as $\lambda_{xy} = 10^{-7}$ is difficult to interpret and may not transfer across experiments.

9.2 Target ratio formulation

The current approach is to choose regularization strength in terms of target ratios relative to the data loss. Define weighted regularization contributions

$$W_{xy} = \lambda_{xy} \mathcal{L}_{xy}, \quad (47)$$

$$W_t = \lambda_t \mathcal{L}_t. \quad (48)$$

Instead of choosing λ_{xy} directly, specify a target ratio

$$\rho_{xy} \approx \frac{W_{xy}}{\mathcal{L}_{\text{data}}} = \frac{\lambda_{xy} \mathcal{L}_{xy}}{\mathcal{L}_{\text{data}}}, \quad (49)$$

and similarly

$$\rho_t \approx \frac{W_t}{\mathcal{L}_{\text{data}}} = \frac{\lambda_t \mathcal{L}_t}{\mathcal{L}_{\text{data}}}. \quad (50)$$

A typical current choice is

$$\rho_{xy} = 0.30, \quad (51)$$

$$\rho_t = 0.30. \quad (52)$$

This means the desired loss balance is approximately

$$\mathcal{L} \approx \mathcal{L}_{\text{data}} + 0.30 \mathcal{L}_{\text{data}} + 0.30 \mathcal{L}_{\text{data}} \quad (53)$$

$$= 1.60 \mathcal{L}_{\text{data}}. \quad (54)$$

The data fraction of the total objective is then

$$\frac{\mathcal{L}_{\text{data}}}{\mathcal{L}} \approx \frac{1}{1 + \rho_{xy} + \rho_t} \quad (55)$$

$$= \frac{1}{1 + 0.30 + 0.30} \quad (56)$$

$$= 0.625. \quad (57)$$

Thus, the model remains primarily data-driven, but the regularization is strong enough to influence the interpolation.

9.3 Reference floor for stability

A fully adaptive scheme based directly on the instantaneous data loss was tested and found to be unstable. When $\mathcal{L}_{\text{data}}$ becomes extremely small, ratios involving $\mathcal{L}_{\text{data}}$ can become numerically unstable, causing weights to oscillate or collapse.

The current solution is to use a stable reference data loss:

$$\mathcal{L}_{\text{ref}} = \max(\text{EMA}(\mathcal{L}_{\text{data}}), \epsilon_{\text{data}}). \quad (58)$$

Here $\text{EMA}(\mathcal{L}_{\text{data}})$ is an exponential moving average of the data loss, and ϵ_{data} is a floor. The floor is not a formal measurement uncertainty. It is an engineering tolerance that prevents the data loss from becoming an unstable denominator.

A corresponding effective target for the weighted spatial term is

$$W_{xy}^* = \rho_{xy} \mathcal{L}_{\text{ref}}, \quad (59)$$

and for the temporal term,

$$W_t^* = \rho_t \mathcal{L}_{\text{ref}}. \quad (60)$$

If smoothed estimates of the raw curvature losses are denoted by $\widehat{\mathcal{L}}_{xy}$ and $\widehat{\mathcal{L}}_t$, then the effective weights can be estimated as

$$\lambda_{xy}^* = \frac{\rho_{xy} \mathcal{L}_{\text{ref}}}{\widehat{\mathcal{L}}_{xy} + \delta}, \quad (61)$$

$$\lambda_t^* = \frac{\rho_t \mathcal{L}_{\text{ref}}}{\widehat{\mathcal{L}}_t + \delta}, \quad (62)$$

where δ is a small numerical stabilizer. The exact implementation may smooth or ramp these weights, but this is the conceptual idea.

9.4 Regularization ramp

The regularization weights are ramped up during early training. If s is the training step and S_{ramp} is the ramp length, then a simple linear ramp is

$$r(s) = \min\left(1, \frac{s}{S_{\text{ramp}}}\right). \quad (63)$$

The effective loss can then be written as

$$\mathcal{L}(s) = \mathcal{L}_{\text{data}} + r(s)\lambda_{xy}\mathcal{L}_{xy} + r(s)\lambda_t\mathcal{L}_t. \quad (64)$$

This gives the model an initial opportunity to fit the data before the curvature priors become fully active.

10 Temporal windows

10.1 Motivation

A single global model over the entire radar interval is not physically ideal because the plasma morphology is not expected to remain correlated over arbitrarily long times. The working assumption is that morphology is meaningfully correlated on approximately one-hour scales, not necessarily over the full multi-hour experiment.

Therefore, the data are divided into overlapping temporal windows. Each window is trained independently:

$$f_{\theta_k}(x, y, t) \approx \log_{10} N_e(x, y, t), \quad t \in [t_k^{\text{start}}, t_k^{\text{end}}]. \quad (65)$$

The subscript k denotes the window index.

10.2 Window size in records and physical time

If the radar cadence is Δt_{rec} , and a window contains N_{win} records, then the physical window duration is approximately

$$T_{\text{win}} \approx N_{\text{win}} \Delta t_{\text{rec}}. \quad (66)$$

For the 5 minute integration product, the earlier choice was

$$N_{\text{win}} = 11, \quad (67)$$

$$\Delta t_{\text{rec}} = 5 \text{ min}, \quad (68)$$

$$T_{\text{win}} \approx 55 \text{ min}. \quad (69)$$

For the 2 minute integration product, the equivalent choice is

$$N_{\text{win}} = 31, \quad (70)$$

$$\Delta t_{\text{rec}} = 2 \text{ min}, \quad (71)$$

$$T_{\text{win}} \approx 62 \text{ min}. \quad (72)$$

The record count changes with cadence, but the physical duration is kept close to one hour.

10.3 Window stride

Let S_{rec} be the stride in records. If window k starts at record index a_k , then

$$a_k = a_0 + kS_{\text{rec}}. \quad (73)$$

For the 2 minute product, a typical choice is

$$S_{\text{rec}} = 15, \quad (74)$$

which corresponds to a new window every

$$15 \times 2 \text{ min} = 30 \text{ min}. \quad (75)$$

This gives overlap between neighboring windows. A 31 record window has a half-width of approximately 15 records, so a stride of 15 records places the next window center near the previous window edge. This gives roughly two-window coverage for interior times.

10.4 Number of windows

If the full file contains N_{rec} records, then the last valid start index is

$$a_{\text{max}} = N_{\text{rec}} - N_{\text{win}}. \quad (76)$$

The number of valid stride-aligned windows is

$$N_{\text{windows}} = \left\lfloor \frac{N_{\text{rec}} - N_{\text{win}}}{S_{\text{rec}}} \right\rfloor + 1. \quad (77)$$

For example, for a 2 minute file with

$$N_{\text{rec}} = 1122, \quad (78)$$

$$N_{\text{win}} = 31, \quad (79)$$

$$S_{\text{rec}} = 15, \quad (80)$$

we get

$$N_{\text{windows}} = \left\lfloor \frac{1122 - 31}{15} \right\rfloor + 1 \quad (81)$$

$$= \lfloor 72.733 \dots \rfloor + 1 \quad (82)$$

$$= 73. \quad (83)$$

The valid starts are therefore

$$0, 15, 30, \dots, 1080. \quad (84)$$

10.5 Record stride versus window stride

It is important to distinguish two different concepts:

- **Record stride inside a dataset window:** this subsamples records within a window. A value of 1 means use every radar record in the window.
- **Window stride:** this controls how far the next window start moves in the full file.

For the 2 minute product, the current desired behavior is usually

$$\text{record stride inside window} = 1, \quad (85)$$

$$\text{window stride} = 15. \quad (86)$$

This means each model uses all 31 consecutive radar records in its local window, while consecutive trained windows start 15 records apart.

11 Per-window prediction and plotting

11.1 Current diagnostic philosophy

At the current stage, windows are not merged. Each trained window is inspected independently. This is intentional. Before defining a continuous merged product, it is necessary to verify that individual window models produce plausible fields for every measured time step within the window.

For each trained window, the diagnostic plotting now evaluates the model at all available time steps where real radar data exist. The measured radar points are overlaid on the predicted continuous field. This lets us inspect whether the model agrees with real data and whether interpolation between beams appears physically reasonable.

11.2 Prediction grid

For a fixed time t_j , the model is evaluated on a grid of horizontal coordinates:

$$\{(x_m, y_n, t_j)\}_{m,n}. \quad (87)$$

The predicted field is

$$\hat{u}_{mn}(t_j) = f_\theta(x_m, y_n, t_j). \quad (88)$$

After denormalization, the plotted field is

$$\widehat{\log_{10} N_{emn}}(t_j) = \text{denorm}(\hat{u}_{mn}(t_j)). \quad (89)$$

Measured points at the same time are overlaid as

$$\{(x_i, y_i, u_i) : t_i = t_j\}. \quad (90)$$

11.3 Color scale and colormap

The current visualization uses the plasma colormap. For comparisons across time or across windows, a fixed color range may eventually be preferable. For single-window inspection, a per-window color range can help reveal structure, but it makes cross-window visual comparison less direct.

12 Comparison between 5 minute and 2 minute integration products

12.1 Motivation

Two large trained runs now exist: one based on a 5 minute integration product and one based on a 2 minute integration product. The 5 minute product has lower temporal cadence but likely better statistical stability per record. The 2 minute product has higher temporal cadence but may have noisier estimates.

The INR framework allows both products to be represented as continuous fields:

$$f_{\theta}^{5\min}(x, y, t) \approx \log_{10} N_e^{5\min}(x, y, t), \quad (91)$$

$$f_{\theta}^{2\min}(x, y, t) \approx \log_{10} N_e^{2\min}(x, y, t). \quad (92)$$

12.2 Difference field

A direct comparison can be defined as

$$\Delta f(x, y, t) = f_{\theta}^{2\min}(x, y, t) - f_{\theta}^{5\min}(x, y, t). \quad (93)$$

Since both are in $\log_{10} N_e$, a difference Δf corresponds to a multiplicative density ratio:

$$\frac{N_e^{2\min}}{N_e^{5\min}} = 10^{\Delta f}. \quad (94)$$

For small Δf , the approximate fractional difference is

$$\frac{N_e^{2\min} - N_e^{5\min}}{N_e^{5\min}} \approx \ln(10)\Delta f. \quad (95)$$

This provides a physically interpretable way to convert log-density differences into approximate percentage differences.

12.3 Useful metrics

Possible quantitative diagnostics include

$$\text{RMSE}_{\Delta} = \sqrt{\frac{1}{M} \sum_{m=1}^M (\Delta f_m)^2}, \quad (96)$$

$$\text{MAE}_{\Delta} = \frac{1}{M} \sum_{m=1}^M |\Delta f_m|, \quad (97)$$

$$P_{95,\Delta} = \text{percentile}_{95}(|\Delta f_m|). \quad (98)$$

These comparisons should be computed both on grids and at actual radar measurement locations.

13 Window merging: next methodological step

13.1 Problem statement

After training overlapping windows, a given physical time may be covered by multiple window models. Suppose the models that cover time t are indexed by $k \in \mathcal{K}(t)$. Each model produces a prediction:

$$f_{\theta_k}(x, y, t), \quad k \in \mathcal{K}(t). \quad (99)$$

The goal is to define a merged field

$$\hat{f}(x, y, t) = \text{merge} \left(\{f_{\theta_k}(x, y, t)\}_{k \in \mathcal{K}(t)} \right). \quad (100)$$

13.2 Nearest-center selection

The simplest option is to use the window whose center time is closest to t :

$$k^*(t) = \arg \min_{k \in \mathcal{K}(t)} |t - c_k|, \quad (101)$$

$$\hat{f}(x, y, t) = f_{\theta_{k^*(t)}}(x, y, t). \quad (102)$$

This is simple and avoids averaging inconsistent models, but it can introduce discontinuities when the selected window changes.

13.3 Weighted averaging

A smoother option is weighted averaging:

$$\hat{f}(x, y, t) = \frac{\sum_{k \in \mathcal{K}(t)} w_k(t) f_{\theta_k}(x, y, t)}{\sum_{k \in \mathcal{K}(t)} w_k(t)}. \quad (103)$$

The weights should be largest near the center of each window and smaller near the window edges.

13.4 Gaussian center weighting

One candidate is Gaussian weighting:

$$w_k(t) = \exp \left(-\frac{(t - c_k)^2}{2\sigma_w^2} \right), \quad (104)$$

where c_k is the center time of window k , and σ_w controls how quickly the weight decreases away from the center.

13.5 Triangular center weighting

Another simple option is triangular weighting:

$$w_k(t) = \max\left(0, 1 - \frac{|t - c_k|}{R_w}\right), \quad (105)$$

where R_w is the half-width of the window in time. This gives zero weight at or beyond the window edge and maximum weight at the center.

13.6 Overlap disagreement as uncertainty proxy

Before merging, disagreement between overlapping windows can be used as a practical uncertainty proxy:

$$\mu_f(x, y, t) = \frac{1}{|\mathcal{K}(t)|} \sum_{k \in \mathcal{K}(t)} f_{\theta_k}(x, y, t), \quad (106)$$

$$\sigma_f^2(x, y, t) = \frac{1}{|\mathcal{K}(t)| - 1} \sum_{k \in \mathcal{K}(t)} (f_{\theta_k}(x, y, t) - \mu_f(x, y, t))^2. \quad (107)$$

Large σ_f would indicate that overlapping window models disagree, suggesting lower confidence in the merged reconstruction.

14 Possible future improvements

14.1 Use AMISR measurement uncertainty in the data loss

If measurement uncertainties σ_i are available for $\log_{10} N_{e,i}$, the data loss could become a weighted least-squares objective:

$$\mathcal{L}_{\text{data}}^{\text{weighted}} = \frac{1}{N_w} \sum_{i=1}^{N_w} \frac{(f_{\theta}(x_i, y_i, t_i) - u_i)^2}{\sigma_i^2}. \quad (108)$$

This would reduce the influence of uncertain measurements and increase the influence of reliable measurements. If uncertainty is given in linear density rather than log-density, it must be transformed carefully.

For small relative uncertainty, if $u = \log_{10} N_e$, then

$$\sigma_u \approx \frac{1}{\ln(10)} \frac{\sigma_{N_e}}{N_e}. \quad (109)$$

14.2 Constrained optimization interpretation

The weighted loss can be interpreted as a relaxed constrained optimization problem. Instead of minimizing

$$\mathcal{L}_{\text{data}} + \lambda_{xy}\mathcal{L}_{xy} + \lambda_t\mathcal{L}_t, \quad (110)$$

one could state the goal as

$$\min_{\theta} \quad \mathcal{L}_{xy} + \alpha_t\mathcal{L}_t, \quad (111)$$

$$\text{subject to} \quad \mathcal{L}_{\text{data}} \leq \epsilon. \quad (112)$$

This formulation makes the philosophy explicit: fit the data up to an acceptable tolerance, then prefer the smoothest field. The current weighted-sum approach is the practical implementation, but the constrained view may be useful for explaining the method.

14.3 Improved SIREN variants

The current baseline is low-frequency SIREN. Future work should test improvements that preserve the sinusoidal neural-field idea while improving stability, interpolation quality, or spectral control. The important point is to compare alternatives against the current strong baseline, not to replace the architecture prematurely.

14.4 FINER-style alternatives

FINER-style implicit representations are a natural next comparison because they remain close to the sinusoidal INR family. The motivation is to test whether a more flexible or better-conditioned periodic representation can preserve smooth plasma morphology while improving detail.

14.5 K-Planes and factorized representations

If the project moves to full four-dimensional reconstruction,

$$f_{\theta}(x, y, z, t) \approx \log_{10} N_e(x, y, z, t), \quad (113)$$

then pure coordinate MLPs may become expensive or difficult to train. Factorized representations such as K-Planes may become attractive because they can represent high-dimensional fields with a more efficient structure.

14.6 Grid-based or hybrid representations

A future comparison should include learned-grid or hybrid grid-MLP models. The concern is that pure INRs may not always be the best representation for all signals. However, grid-based methods introduce their own interpolation assumptions and should be compared only after the current physics-aware SIREN baseline is stable.

14.7 Explicit probabilistic or ensemble uncertainty

Uncertainty can be approached in several ways:

- train ensembles with different random seeds;
- use overlap disagreement between windows;
- predict both mean and variance;
- use dropout-style approximate uncertainty;
- incorporate AMISR measurement uncertainty directly in the likelihood.

A possible probabilistic output model is

$$f_{\theta}(x, y, t) = (\mu_{\theta}(x, y, t), \log \sigma_{\theta}^2(x, y, t)), \quad (114)$$

with a Gaussian negative log-likelihood loss

$$\mathcal{L}_{\text{NLL}} = \frac{1}{N} \sum_{i=1}^N \left[\frac{(u_i - \mu_{\theta}(x_i, y_i, t_i))^2}{2\sigma_{\theta}^2(x_i, y_i, t_i)} + \frac{1}{2} \log \sigma_{\theta}^2(x_i, y_i, t_i) \right]. \quad (115)$$

This is more complex and should likely come after the deterministic baseline and window-merging approach are established.

15 Implementation status

The main implementation improvements so far are:

1. The dataset reads directly from AMISR HDF5 files.
2. The model input is now (x, y, t) , not only (x, y) .
3. The training is performed on local temporal windows.
4. The derivative regularization uses second derivatives rather than first derivatives.
5. The spatial loss is a horizontal Hessian penalty.
6. The temporal loss is an f_{tt}^2 penalty.

7. The derivative losses are evaluated at collocation points.
8. Low-frequency SIREN settings are used because they produce more plausible plasma morphology.
9. Raw lambda sweeps were performed and then replaced by a more interpretable target-ratio strategy.
10. A reference-based loss scale was introduced to avoid instability when data loss becomes very small.
11. Scripts now support many independent windows in large output folders.
12. Additional plotting scripts generate predictions for every real time step in every trained window.

16 Key decisions to explain in the meeting

16.1 Decision 1: Low-frequency SIREN

The use of low-frequency SIREN is a domain-driven architectural decision. Higher-frequency SIREN can fit sparse data but produces visually implausible plasma structure. Lower-frequency SIREN acts as a spectral prior and gives fields that better match plasma-physics expectations.

16.2 Decision 2: Curvature instead of gradient regularization

The regularization is not a gradient penalty. It is a second-derivative penalty. This is important because plasma can have large-scale gradients. The method should allow gradients but suppress unsupported rapid bending.

16.3 Decision 3: Horizontal Hessian, not just Laplacian

The spatial regularization is better described as a horizontal Hessian curvature penalty:

$$\mathcal{L}_{xy} = \mathbb{E} [f_{xx}^2 + 2f_{xy}^2 + f_{yy}^2]. \quad (116)$$

This is different from a Laplacian penalty:

$$\mathcal{L}_{\Delta} = \mathbb{E} [(f_{xx} + f_{yy})^2]. \quad (117)$$

The Hessian penalty controls curvature in any horizontal direction.

16.4 Decision 4: Local time windows

The plasma morphology is not assumed to remain correlated across the full experiment. Therefore, the reconstruction is trained in approximately one-hour windows. This is physically motivated and

prevents a single global network from imposing long-range temporal continuity that may not exist.

16.5 Decision 5: Target ratios instead of raw lambdas

Raw λ values are hard to interpret and may not transfer across windows. Target ratios are easier to explain:

$$\frac{\lambda_{xy}\mathcal{L}_{xy}}{\mathcal{L}_{\text{data}}} \approx 0.30, \quad (118)$$

$$\frac{\lambda_t\mathcal{L}_t}{\mathcal{L}_{\text{data}}} \approx 0.30. \quad (119)$$

This says that the regularization terms are substantial but still secondary to the data term.

16.6 Decision 6: Do not merge windows yet

The current inspection stage treats windows independently. This avoids hiding problems behind a merging procedure. Window merging is the next step, but first the individual window models must be trusted.

17 Questions for Gonzalo Ruz

The following questions are suggested for discussion:

1. Is the horizontal Hessian curvature penalty the right regularizer, or should we use a different smoothness prior?
2. Should the method be framed as regularized empirical risk minimization, constrained optimization, or a weak PINN-like method?
3. Is the target-ratio loss balancing strategy defensible, or should we use a more formal multi-objective loss balancing approach?
4. Should the regularization ratios be constant across all windows, or should they vary based on measurement noise or local plasma structure?
5. What is the best first approach to merging overlapping windows: nearest-center selection, triangular weighting, Gaussian weighting, or learned/error-weighted blending?
6. Can overlap disagreement be used as a meaningful uncertainty proxy?
7. Should measurement uncertainty from AMISR be incorporated directly into the data loss?
8. At what point should we compare against FINER, K-Planes, or grid-based hybrid representations?
9. Does low-frequency SIREN provide a sufficient spectral prior, or should the frequency content be explicitly constrained or learned?

10. For future 4D reconstruction, should altitude be treated as another coordinate, or should separate altitude bands remain preferable for interpretability?

18 One-paragraph meeting summary

We are using a low-frequency SIREN implicit neural representation to reconstruct $\log_{10} N_e(x, y, t)$ from sparse AMISR measurements. The model is trained on overlapping temporal windows of approximately one hour, motivated by the limited physical correlation time of ionospheric plasma morphology. The loss combines a data-fitting term with second-derivative regularization in space and time. The spatial regularization penalizes the Frobenius norm of the horizontal Hessian, while the temporal term penalizes f_{tt}^2 . This allows large-scale plasma gradients while suppressing unphysical curvature and oscillatory artifacts between radar beams. Instead of manually tuning raw regularization weights, the current implementation uses target loss ratios relative to a reference data loss, with a floor for numerical stability. The immediate next steps are to validate all individual windows, compare the 5 minute and 2 minute integration products, and define a principled method for merging overlapping window predictions into a continuous time-dependent reconstruction.

19 Compact list of equations

For convenience, the main equations are collected here.

$$f_\theta(x, y, t) \approx \log_{10} N_e(x, y, t), \quad (120)$$

$$\mathbf{r}_i = \begin{bmatrix} \tilde{x}_i \\ \tilde{y}_i \\ \tilde{t}_i \end{bmatrix}, \quad (121)$$

$$\mathbf{h}_\ell = \sin(\omega_\ell (W_\ell \mathbf{h}_{\ell-1} + \mathbf{b}_\ell)), \quad (122)$$

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{xy} \mathcal{L}_{xy} + \lambda_t \mathcal{L}_t, \quad (123)$$

$$\mathcal{L}_{\text{data}} = \frac{1}{N_w} \sum_{i=1}^{N_w} (f_\theta(\tilde{x}_i, \tilde{y}_i, \tilde{t}_i) - \tilde{u}_i)^2, \quad (124)$$

$$H_{xy} f_\theta = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}, \quad (125)$$

$$\mathcal{L}_{xy} = \mathbb{E} [f_{xx}^2 + 2f_{xy}^2 + f_{yy}^2], \quad (126)$$

$$\mathcal{L}_t = \mathbb{E} [f_{tt}^2], \quad (127)$$

$$\nabla_{xy}^2 f_\theta = f_{xx} + f_{yy}, \quad (128)$$

$$\rho_{xy} = \frac{\lambda_{xy} \mathcal{L}_{xy}}{\mathcal{L}_{\text{data}}}, \quad (129)$$

$$\rho_t = \frac{\lambda_t \mathcal{L}_t}{\mathcal{L}_{\text{data}}}, \quad (130)$$

$$\mathcal{L}_{\text{ref}} = \max(\text{EMA}(\mathcal{L}_{\text{data}}), \epsilon_{\text{data}}), \quad (131)$$

$$N_{\text{windows}} = \left\lfloor \frac{N_{\text{rec}} - N_{\text{win}}}{S_{\text{rec}}} \right\rfloor + 1, \quad (132)$$

$$\hat{f}(x, y, t) = \frac{\sum_{k \in \mathcal{K}(t)} w_k(t) f_{\theta_k}(x, y, t)}{\sum_{k \in \mathcal{K}(t)} w_k(t)}, \quad (133)$$

$$w_k(t) = \exp\left(-\frac{(t - c_k)^2}{2\sigma_w^2}\right), \quad (134)$$

$$\Delta f(x, y, t) = f_\theta^{2\min}(x, y, t) - f_\theta^{5\min}(x, y, t), \quad (135)$$

$$\frac{N_e^{2\min}}{N_e^{5\min}} = 10^{\Delta f}. \quad (136)$$

20 Closing statement

The project is no longer a simple demonstration that SIREN can fit radar data. It has become a structured approach to plasma-field reconstruction from sparse measurements. The key idea is that the machine-learning model should not be allowed to invent arbitrary structure. The combination of low-frequency SIREN, local temporal windows, second-derivative curvature priors, target-ratio loss balancing, and conservative diagnostics creates a defensible baseline for moving toward a continuous, physically interpretable reconstruction of ionospheric plasma density.